Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/02/2026
PI Name:	Lewis, Norman G Ph.D.		
Project Title:	Dissecting Beneficial Plant-Microbe Interactions and Their Efficacy in the ISS Spaceflight Environment, a Model Study		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	99164-6340	**Congressional District:**	5
Comments:			
Project Type:		**Solicitation / Funding Source:**	03-OBPR-02
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per NSSC information (Ed, 1/3/25) NOTE: End date changed to 12/31/2024 per NSSC information (Ed, 3/12/24) NOTE: End date changed to 12/31/2023 per NSSC information (Ed, 1/15/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davin, Laurence Ph.D. (Washington State University, Pullman) Kahn, Michael Ph.D. (Washington State University, Pullman)		
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Task Description:	Significance of objectives to NASA and this Solicitation: Deep space exploration or extraterrestrial colonization (e.g., Moon or Mars) will require the ability to sustainably produce plants for human/animal consumption, as well as providing aesthetic benefits of plant life to future crews and personnel in extra-terrestrial colonies. One key challenge in spaceflight/microgravity is in overcoming long-standing difficulties in efficaciously providing water and nutrients to germinating and maturing plants. Another important research challenge that has seen little attention is in productively exploiting beneficial plant-microbe interactions in spaceflight/microgravity, particularly for nitrogen (N) fixation. When both challenges are resolved for optimal, productive, and efficacious plant growth in space, this will provide the exciting opportunity to recycle organically bound carbon (C) and N that was sequestered in these plants. Through subsequent recycling of those organics (e.g., derived from human and animal consumption waste and from unused plant parts), this will help enable sustainable plant growth over multiple generations. Another benefit of studying beneficial plant microbe interactions is at the fundamental science level, i.e., by gaining much improved understanding of how the spaceflight/microgravity environment affects this important physiological process. Central objectives of proposed research: Complementary purposes of our 2 Specific Aims are to initially dissect, understand, and optimize plant growth/development in spaceflight/microgravity via exploiting beneficial plant-microbe interactions. Then to ultimately recycle organic C and N from them suitable for subsequent multiple plant generations. To do this, we will use model Medicago plant species (e.g., alfalfa), and its beneficial bacterial symbiont, which together can potentially displace the need for N-containing fertilizer in spaceflight/microgravity. Specific Aims: 1. Comprehensively compare and contrast efficacy of beneficial symbiotic plant-microbe interactions between Medicago and Sinorhizobium growing on the International Space Station (ISS) and on Earth (NASA Kennedy Space Center/KSC), including understanding changes occurring at the molecular level; 2. Compare and contrast ease of biodegradation of the ISS and Earth grown Medicago plant material, in order to assess whether there are any differences in the ability to recycle C, N, etc. for multiple generations of plant growth/development. Justification for this work is threefold: The first is to demonstrate that beneficial plant microbe interactions during N-fixation can be efficaciously achieved in spaceflight/microgravity. The second is to gain a fundamental understanding of spaceflight/microgravity environment effects on these beneficial plant microbe interactions, and their potential usage for deep space exploration and colonization. The third is to demonstrate that organic C and N can be facilely recycled to support multiple generations of plant growth in space. Methods/techniques: During growth, plant size measurements will be carried out to assess comparative N-fixation efficacy for each condition, both on the ISS and on Earth. Tissues (leaf, stem, and root) from the ISS and Earth control will be collected after ca. 6-8 weeks growth, frozen (-160°C). They will be subjected to transcriptomic and metabolomics (including amino acid) analyses; the microbiomes present in aerial/underground tissues will be determined. The multi-omics approaches employed are as for our Arabidopsis study. Medicago plant material, from the ISS and Kennedy Space Center (KSC) ground control, will also be subjected to biodegradation to establish whether there are any differences in N-mineralization (for recycling) in spaceflight/microgravity or ground control tissues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Among the benefits on Earth envisaged: improving our knowledge of N-fixing process and the symbiosis between Medicago and Sinorhizobium, and determining optimal lignin contents for space and Earth will be very instructive, as will the recycling C/N capabilities for both wild type and genetically modified plant lines. Demonstrating this in space is also a very effective means of demonstrating to aspiring young scientists (including Middle and High School students) and others of the importance of plant life, of N-fixation, and of C/N recycling in a sustainable manner.
	1. Experiment Verification Test An Experiment Verification Test (EVT) was carried out June - July 2024. Low lignin (LL) alfalfa (Medicago sativa), its corresponding wild type (WT), and Ladak alfalfa were grown from seed in two Veggie (1 and 2) units, each containing 12 plant growth systems (PGSs). To each PGS was added 2 wicks with 6 seeds/wick. On average, 81 and 84% of the seeds germinated in Veggie 1 and 2, respectively. All germinated plants continued to grow normally, with the Ladak/S. meliloti forming numerous nodules in all 6 PGSs. All success criteria were met and were in the Excellent sub-category. [Ed. Note: The Vegetable Production System (Veggie) is a plant growth unit on the International Space Station. It is capable of producing salad-type crops to provide the crew with a safe, palatable, and nutritious source of food, while also providing a tool to support recreation and relaxation.] 2. Additional Testing: Elmer’s glue versus guar gum Seed adhesion to the germination paper originally used guar gum in the successful EVT. However, because of the somewhat irregular shapes and sizes of alfalfa seeds, adhesion was, in some instances, poorer – such seeds showed a tendency to get dislodged. In order to overcome that issue, the International Space Station (ISS) flight-approved Elmer’s glue was also tested in 2024 as a more reliable and better adhesive for alfalfa seeds to the germination paper. Four PGSs were used for this compatibility/comparison test, with seeds glued on the germination paper either with Elmer’s glue (2 PGSs) or guar gum (2 PGSs). Two of the PGSs (1 with Elmer’s glue, one with guar gum) had S. meliloti applied to the germination paper. [This was to ensure Elmer’s glue did not inhibit germination.] Germination was initiated by addition of nutrient solution, with the alfalfa plants 42 days after initiation (DAI). Germination rate observed was 75/92% with Elmer’s glues and 83/92% with guar gum. Biomass yields were comparable. Nodules were present in large numbers. Elmer’s glue was demonstrated to be a much more reliable and better adhesive for adhering alfalfa seeds to the germination paper. 3. Additional Testing: Length of storage time before initiation Further testing was requested by NASA Kennedy Space Center (KSC) personnel in January 2025 to ensure reliability of our experimental design should there be an extended storage time on ISS. Experiment initiation was carried out by addition of nutrient solution (140 ml) with or without S. meliloti to each PGS. KSC personnel asked to test S. meliloti

Task Progress:	viability if initiation was carried out up to 3 months (12 weeks) after liftoff. Thus, 4 PGSs were set-up, with initiation carried out at 0, 4, 8, and 12 weeks. Nodules were formed in each case. 4. Unexpected NASA designated sub-contractor replacement for support of our study The EVT was satisfactorily completed in July 2024 with support personnel from Aegis, with our additional requisite work including American Society for Gravitational and Space Research (ASGSR) presentations and Progress Report completed by December 2024. NASA initiated a new sub-contractor competition with applications reviewed by NASA in Spring 2025. - Replacement of Aegis personnel at KSC by Leidos personnel at the NASA Johnson Space Center (JSC) then occurred. Everything now is hopefully in place to conduct the long-awaited ISS study using a combination of support personnel at KSC and JSC. 5. Progress towards completion of the ERD for consideration by BPS A final version of the ERD was completed by 14 November 2025 and was shared with NASA Biological and Physical Sciences (BPS) personnel for consideration for approval to complete our ISS investigation. We had hoped our study would have already been flown this year. 6. Additional Testing: KSC requested “mock-up experiment” of our Veggie-06 study and vibration testing in November 2025 The latter was to: a) conduct a vibration test to compare and contrast Elmer’s glue vs guar gum suitability for seed adherence; and b) initially help provide oversight for an in-person experience at KSC for Leidos personnel located at JSC as regards the Veggie-06 experimental design and execution. Progress made was as follows: a) KSC personnel requested that we conduct a mock setup of Veggie-06 again using ISS-approved Elmer’s glue and guar gum for side-by-side comparison, with the addition of a vibration test. The Veggie-06 mockup was done on October 28, 2025 by Jeff Richards (LASSO), Rachel Tucker, and Tom Dreschel (LASSO) with our step-by-step oversight and advisement done remotely in real time; Jeff Richards prior to that had conducted the needed successful alfalfa seed sterilization and germination procedures, and he and Rachel Tucker together with Tom Dreschel set up the Veggie-06 hardware on October 30. The vibration test on November 4 resulted in expected results: the Elmer’s glue kept the alfalfa seed adhered to the germination paper, whereas the alfalfa seeds adhered to the germination paper by guar gum all dislodged. The vibration test thus demonstrated that the ISS flight-approved Elmer’s glue for alfalfa seed adherence was again successful. b) To our initial understanding, the 2-week subsequent test with Veggie-06 hardware to establish germination and growth was to be conducted by Leidos personnel, with our real-time oversight and input of operations remotely. This was to allow Leidos personnel to gain a better understanding of the experimental design and execution. The KSC plan changed. Leidos personnel observed these operations remotely. 7. Progress in genetic manipulation of alfalfa lignin levels This work entailed examining the currently used (newer) alfalfa seed batches of control and low lignin genotypes that were greenhouse grown. Plants were harvested at 6 weeks after germination, with internodes 1 – 6 representing the youngest to the oldest internodes in the stems individually collected. For each genotype, 10 stems were collected, internodes dissected and combined (i.e., 1, 2, …6). Internodes were subjected to lignin monomeric composition analyses: i.e., their estimated lignin levels and guaiacyl (G) and syringyl (S) compositions, the latter two differing in the number of aromatic methoxy groups. The estimated lignin levels were reduced in average by 11%, but also with near 2.5× reduction in S units in the low lignin genotype; these genotypes will be tested for their ease of biodegradability/recycling, e.g., to release monomeric sugars in future work.
Bibliography Type:	Description: (Last Updated: 01/13/2026)
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